Naveen Morla

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PROFESSIONAL SUMMARY

Results-driven **Data Scientist** with 5+ years of experience building end-to-end **ML pipelines, LLM-powered applications,** and **scalable APIs** across manufacturing, legal, and analytics domains. Deep expertise in **RAG systems, MLOps, FastAPI,** and multi-provider AI architectures (Claude, Gemini, DeepSeek) with production deployments on **AWS** and **Azure.**

WORK EXPERIENCE

Data Scientist, Business Intelligence – The Lisinski Law Firm, LLC **June 2025 – Present**

- Developed and maintained **forecasting models** to predict **case growth** and **business development** trends across 50,000+ active cases, leveraging advanced statistical and **machine learning** techniques to support strategic decision-making.
- Conducted **sentiment analysis** on client and case data using **LLM models** to identify key themes and insights, informing **business development** strategies and improving client experience outcomes.
- Applied extensive **immigration knowledge** and **workflow expertise** to analyze case pipelines, identify operational bottlenecks, and deliver data-driven recommendations aligned with immigration processes.
- Built and deployed scalable **data pipelines** and analytics solutions using **Azure Functions** and **Python**-based data science tools, automating reporting workflows and enabling real-time business insights.
- Designed and maintained interactive **Power BI** dashboards to visualize KPIs, case metrics, and business performance, enabling 15+ leadership stakeholders to make informed data-driven decisions across the firm.
- Engineered **high-performance APIs** using **FastAPI** to deliver fast and scalable data endpoints, optimizing response times and supporting real-time analytics consumption across business intelligence applications.
- Designed and implemented end-to-end **CI/CD pipelines** for automated deployment of data science models and APIs to **Azure**, streamlining release cycles and ensuring reliable, repeatable deployments across environments.

Data Scientist / Software Engineer, THK Manufacturing of America, Inc. **Jan 2024 – Jun 2025**

- Engineered high-performance APIs using **FastAPI** with advanced caching for seamless ingestion, processing, and retrieval of machine and inventory datasets; fine-tuned predictive models with **PyTorch** and **TensorFlow** to improve lead-time forecasting accuracy by 18% and optimize inventory management across 5,000+ SKUs.
- Executed comprehensive data preprocessing, feature engineering, and data wrangling to improve model accuracy and robustness, leveraged **Polars** and **PySpark** for accelerated data processing, ensuring efficient handling of large datasets.
- Architected and managed containerized **microservices** with **Kubernetes** and **Docker** for scalable big data processing pipelines handling 1TB+ of daily manufacturing data, implementing **MLOps** practices to streamline the deployment and monitoring of machine learning models.
- Created dynamic data visualizations using **Tableau** for data-driven decision-making and developed responsive frontend interfaces with **React** and **JavaScript** to enhance user experience and data visualization.
- Utilized **SQL** for complex backend data management and querying, implemented **Git** version control, and established **CI/CD** pipelines for streamlined code management, automated testing, and continuous deployment.

Data Scientist, Pago Analytics, India **May 2019 – Apr 2021**

- Utilized **Python**, and **deep learning** libraries to design resume **parsing** techniques to build a Talent Acquisition Management System (TAMS). Developed advanced **AI** features with 30% more efficiency in the hiring process.
- Engineered an automated hiring process and facilitated its deployment using **AWS Sagemaker**. Collected 10,000+ resumes and labeled them automatically and manually for the training data, using libraries like **NLTK, NLP, and Spacy** to analyze data.
- Normalized **SQL** databases, improving data handling and system performance by 2X. Coordinated with cross-functional teams throughout the development and deployment process, following **Agile** methodologies.
- Applied strong analytical skills and creativity in data mining and defining essential **KPIs/metrics**. Communicated complex data points in a comprehensible manner to stakeholders using **Power BI**. Fostered lasting relationships, ensuring smooth project execution, stakeholder satisfaction, and participation in code review.

Data Analyst, Knowledge Matrix Pvt Ltd, India **Mar 2018 – Feb 2019**

- Applied a structured data analysis workflow and built an **ETL pipeline** using tools like Python and SQL to gather, clean, and transform data, uncovering valuable insights for informed decision-making, and saving 5+ hours for 10 million records.
- Utilized popular tools like **Tableau** and **PowerBI** to create visually appealing dashboards and reports, effectively presenting key findings to stakeholders and supporting strategic planning.
- Applied **statistical techniques** and data visualization to identify patterns, correlations, and anomalies, providing actionable insights and recommendations for process improvement and business growth.

PROJECTS

Job Tracker | FastAPI, PostgreSQL, Streamlit, APScheduler, GitHub Actions, Nginx **2024 – Present**

- Built a full-stack job tracking platform that auto-scrapes 30+ company career portals hourly using Python, persisting 10,000+ postings in PostgreSQL with automatic cleanup of records older than 7 days.
- Engineered FastAPI backend with rate limiting, pagination, and RESTful APIs; implemented CI/CD pipelines via GitHub Actions and deployed on Oracle Cloud Infrastructure with Nginx reverse proxy for production access.
- Developed interactive Streamlit dashboard with real-time filtering (role, company, location), dark mode toggle, mobile-responsive layout, and a system monitoring interface tracking CPU, memory, and disk usage.

Auto-Scrapper | FastAPI, LangChain, Playwright, BeautifulSoup, Docker, Gemini, GPT-4 **2024**

- Developed an LLM-powered intelligent web scraping tool that auto-generates and iteratively refines scraping code from natural language descriptions, supporting GPT-4, Claude, and Gemini as interchangeable backends via LangChain.
- Integrated Gemini 2.0 Flash for automated website structure analysis; implemented sandboxed code execution with real-time token usage tracking and interactive status updates on a FastAPI + Bootstrap frontend.
- Containerized the full application with Docker for consistent cross-environment deployment; designed a modular multi-provider architecture with automatic error recovery and code refinement loops.

RAG-LLM (Loan Document Processing) | AWS Bedrock, ChromaDB, LangChain, DeepSeek, Claude **2024**

- Architected a Retrieval-Augmented Generation (RAG) system for intelligent loan document Q&A using AWS Bedrock Titan embeddings for semantic vector search over PDF and CSV documents stored in ChromaDB.
- Implemented multi-LLM support (DeepSeek, GPT-4, Claude-3) with automatic fallback chains, exponential backoff retries, and full source reference tracking for auditability of AI-generated responses.
- Engineered a scalable document ingestion pipeline with embedding caching, batch processing optimization, and progress tracking, reducing repeated computation overhead and accelerating large file processing.

GNN – Graph Neural Network Anomaly Detection | Python, PyTorch, GNN **2023**

- Developed a Graph Neural Network-based anomaly detection system for multivariate time series, leveraging graph-structured representations to model complex inter-sensor dependencies for industrial IoT datasets.
- Designed graph construction from sensor correlation matrices and trained GNN models to identify anomalous temporal patterns, outperforming traditional statistical baselines in detection accuracy.
- Open-sourced the implementation achieving 3 GitHub stars and 1 fork, demonstrating practical applicability of graph deep learning for predictive maintenance in manufacturing environments.

Languages: Python, R, JavaScript, SQL.

Data Science Technologies: Machine Learning, Big Data, OpenCV, LLM Agents, RAG, MLOps, A/B testing, Fine-tuning Models.

Packages/Tools/Frameworks: Scikit-Learn, PyG (PyTorch Geometric), NumPy, SciPy, Pandas, NLTK, LangChain, Matplotlib, Keras, Jupyter Notebook, Docker, Kubernetes, Streamlit, Playwright, BeautifulSoup, Power BI, Tableau, PySpark, Polars, ChromaDB, TensorFlow, PyTorch, Git, JIRA, FastAPI, React, AWS SageMaker, Azure Functions, Hugging Face Transformers, BERT.

Databases: MySQL, MongoDB, Cassandra, Elasticsearch, PostgreSQL, DynamoDB, AWS RDS, NoSQL, Vector databases.

Statistics/ML: Statistical Analysis, Predictive Analytics, Statistical Modelling, Linear/Logistic/Ridge/Lasso Regression, K-means Clustering, Classification, Data Mining, Decision Trees, Recommender Systems, Random Forests, XGBoost, LGBM, SVM.

EDUCATION

- **BOWLING GREEN STATE UNIVERSITY, OHIO** **Aug 2021 – May 2023**

  M.S in Data Science

  Coursework: Data Science Programming, Database Management Systems, Regression Analysis, Data Mining, Time Series Analysis, Artificial Intelligence Methods, Probability Theory I, Mathematical Statistics II, Linear and Integer Programming, Data Science Project.

ACCOMPLISHMENTS/CERTIFICATIONS

Certifications:

- Coursera : Python and Machine Learning, Introduction to Data Analytics.
- Udemy : Tableau, PowerBI, Django.
- Data Camp : Introduction to R programming and statistics in spreadsheets and Excel.